

Outline

Use Bayes rule to mechanically solve probability riddles

Monty Hall
Bertrand's box paradox
The Sisters Paradox
Boy Born on Tuesday
Conclusion

Use Bayes rule to mechanically solve probability riddles

You might have heard Bayes rule is useful to answer the question "Is it likely that a person with a positive test actually has the disease?" [\[xkcd\]](#). But did you know Bayes rule can be applied quite easily even if the outcome is not binary, and without memorizing any formula with " $P(A|B)$ " in it?

1. Start by making a table where the **columns are all possible outcomes** of the question you are interested in.
2. Add a row called "prior": Before you observe any information (e.g. the results of a test) what is the probability for each outcome?
3. Add a row called "likelihood": For each possible outcome, assume that it is true for a moment. How likely are our observations to occur in each of those scenarios?
4. Multiply the two rows and scale the result such that it adds up to 100%. The resulting probability distribution is the updated belief of the outcome ("posterior").

Let's use Bayes rule to solve some classic probability riddles!

Monty Hall

The Monty Hall problem is probably the most well known probability puzzle and none of the explanations ever felt really intuitive to me. But it does not have to be intuitive, because we can solve it mechanically using Bayes rule. If you never heard of the Monty Hall problem, here it is:

There are three closed doors. Behind two doors there are goats, behind one there is a car. If you pick the right door you win the car. You pick one door at random, but before you open it, Monty Hall stops you. He opens another door showing you a goat and asks, if you want to switch to the other yet unopened door.

1. The question we are interested in is "Behind which door is the car?". Lets give each outcome a column.

Which door contains car?	Door 1	Door 2	Door 3
--------------------------	--------	--------	--------

2. **Prior:** Before Monty Hall opens a door, each of the door is equally likely to contain a car.

Which door contains car?	Door 1	Door 2	Door 3
Prior	1/3	1/3	1/3

3. Because we prefer no door, **let us pick Door 1** at random. Monty Hall now gives us new information by **opening Door 3 and revealing a goat**. Clearly the prior probabilities must be updated.

Scenario: **Door 1 actually contains the car**. What is the likelihood of Monty Hall opening Door 3? To show us a goat, he can open either Door 2 or Door 3. Assuming he picks at random, the **likelihood of seeing Door 3 being opened is 1/2**.

Scenario: **Door 2** contains the car. The door we picked contains a goat. But he will never open the door we picked, so **the likelihood of seeing Door 3 being opened is one**.

Scenario: **Door 3** contains the car. We will never see Monty Hall open Door 3, when it contains the car. The likelihood of us observing him opening Door 3 is **zero**.

Which door contains car?	Door 1	Door 2	Door 3
Prior	1/3	1/3	1/3
Likelihood of Door 3 being opened	1/2	1	0

i Note that the likelihoods do not have to add up to one.

4. Multiply the two rows together.

Which door contains car?	Door 1	Door 2	Door 3
Prior	1/3	1/3	1/3

Likelihood of Door 3 being opened	1/2	1	0
Prior * Likelihood	1/6	1/3	0

Right off the bat, we can see Door 2 is more probable to contain the car than Door 1 (our current pick). However to get proper probabilities we need to scale the last row to 100%. Do this by dividing each entry by the sum of all entries. In our case the sum is $1/6 + 1/3 = 1/2$ and dividing by $1/2$ means we have to multiply everything by 2.

Which door contains car?	Door 1	Door 2	Door 3
Prior	1/3	1/3	1/3
Likelihood of Door 3 being opened	1/2	1	0
Posterior	1/3	2/3	0

Bertrand's box paradox

This paradox also comes in a more delicious pancake form:

"There are three pancakes: One is burned on both sides, one is burned only on one side. The third pancake is not burned at all. You are served a pancake on a plate. The upper side of the pancake is burned. What is the probability that the other side is also burned?"

Sides burned?	Zero	One	Two
Prior	1/3	1/3	1/3
Likelihood of observing upper side being burned	0	1/2	1
Posterior	0	1/3	2/3

Note that this is the exact same table as the Monty Hall problem.

The Sisters Paradox

"A family has two children. You're told that at least one of them is a girl. What's the probability both are girls?" [\[Source\]](#)

Number of girls?	Zero	One	Two
Prior	1/4	1/2	1/4
Likelihood of at least one girl	0	1	1
Posterior	0	2/3	1/3

Boy Born on Tuesday

"Mrs. Chance has two children of different ages. At least one of them is a boy born on Tuesday. What is the probability that both of them are boys?" [\[Source\]](#)

Number of boys?	Zero	One	Two
Prior	1/4	1/2	1/4
Likelihood of at least one boy born on Tuesday	0	1/7	$1-(6/7)^2$

To calculate the likelihood of at least one boy being born on a Tuesday in the scenario where Mrs. Chance has two boys, consider the negation of this statement and subtract its probability from 100%: What is the likelihood no boy is born on a Tuesday? Not being born on a Tuesday has a probability of 6/7. Assuming independence, the likelihood of both boys not being born on a Tuesday is $(6/7)^2$.

Let us calculate the posterior distribution using **R**:

```
> prior <- c(1/4, 1/2, 1/4)
> likelihood <- c(0, 1/7, 1-(6/7)^2)
> x <- prior * likelihood
> x / sum(x)
[1] 0.0000000 0.5185185 0.4814815
```

Thus, the answer to this riddle is about **48%**.

Conclusion

All of the probability riddles above can be swiftly solved by using Bayes rule. The process of filling out the tables above is quite mechanical and computation of the posterior distribution does not require remembering any formulas beyond "prior times likelihood" and normalizing to 100%.

Bayes rule can do much heavier lifting than answering questions with discrete and finite outcomes. The principle of "prior times likelihood" also works for more interesting questions where the possible outcomes are infinite and multi-dimensional. A truly great lecture on Bayesian data analysis that focuses on the application rather than the mathematics is [\[Statistical Rethinking\]](#) by Richard McElreath.